Shapes and Their Names — Turning a Shape Does Not Change It
Answer Key

Kindergarten / Grade 1

Quick Answers

1. 3, 2, 1	3. 4	5. 3	7. —
2. —	4. 4, 4	6. —	8. —

Curriculum

Level: Kindergarten / Grade 1
K.G / 1.G – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 11 tagged checks.

What this sheet is teaching, and what to listen for: at this age children learn shape names from pictures that are almost always drawn the same way up — triangles resting on a flat bottom, squares with horizontal tops. The result is a very common and very fixable misunderstanding: a square turned onto its corner gets a new name, "diamond," and a triangle standing on its point stops being a triangle. The cure is not to tell the child the rule but to have them *count*. Sides and corners do not change when a shape is turned, and counting is something a five-year-old can do and trust. When you hear "that's a diamond," answer with "how many sides?" rather than "no."

1. Sorting Shape Set One.

Shapes A, D and F are triangles: three straight sides and three corners each, even though A is upside down, D leans over, and F points sideways. Shapes B and E are squares — B is balanced on a corner and E is turned partway round, but both have four equal straight sides and four square corners. Shape C is the circle: one curved edge, no corners at all.

triangles	squares	circles
$\boxed{3}$	$\boxed{2}$	$\boxed{1}$

If the child misses one: it is nearly always A or B — the upside-down triangle and the square on its corner. Cover the rest of the row with your hand, turn the page until that shape looks familiar, and let them name it. Then turn the page back and ask whether it changed.

2. Which are there more of?

From the counts in problem 1, there are 3 triangles and 2 squares, so there are more triangles. Written as a comparison:

$$\boxed{3 > 2}$$

The child should circle **more triangles**.
The point of asking: the comparison can only be made after the shapes have been named, so a child who called shape B a diamond will get 3 and 1 here and still answer "more triangles". The right answer for the wrong reason — worth checking their tally in problem 1 rather than just this circle.

3. Turned every which way.

The first four shapes are all triangles: three straight sides, three corners. One points down, one leans right, one leans left, one points sideways — none of that changes the count. The fifth shape is the rectangle, and it gets the X: four sides, four corners.
The child should have coloured $\boxed{4}$ triangles.
Listen for the reason. "It's not a triangle because it has four sides" is the answer you want. "It's not a triangle because it looks different" means the counting habit has not taken hold yet — go back and count sides on each of the five shapes together.

4. Count them and see.

Have the child touch each straight edge and say a number, then each corner and say a number.
Sides: $\boxed{4}$. Corners: $\boxed{4}$.
The four sides are all the same length and all four corners are square corners, so its name is **square**. It is balanced on a corner, and that is all.
About "diamond": it is not a wrong word so much as a word about *position* rather than *shape*. "Diamond" tells you which way it is turned; "square" tells you what it is. A helpful thing to say: "you can call it a diamond if you like — and it is still a square."

5. Find the rectangles.

The rectangles are the tall one standing on end, the one leaning over, and the wide one lying flat — $\boxed{3}$ altogether. Each has four straight sides and four square corners, with opposite sides the same length.
The other three are the circle (curved, no corners), the triangle (three sides), and the hexagon (six sides).
Worth naming out loud: the three rectangles are not the same size or the same way up, and they are all rectangles anyway. Size, position and turn are all things that can change without the name changing. Sides and corners are what the name is made of.
A square is also a rectangle — true, and not something to raise here unless the child raises it. If they do, say yes, a square is a special rectangle whose sides are all equal, and let it rest there.

6. Which group has more four-sided shapes?

Group A holds a square on its corner, a leaning rectangle, a triangle and nothing else with four sides — so two four-sided shapes.
Group B holds a square, a tall rectangle, a square on its corner and a leaning rectangle — four four-sided shapes.
Comparing the two counts:

$$\boxed{2 < 4}$$

so Group B has more.
Why this problem is harder than problem 2: the child has to apply a *rule* (four straight sides) rather than recall a *name*, and the rule has to survive the shapes being turned. A child who counts three in Group A has counted the triangle; a child who counts three in Group B has missed one of the turned ones.

7. The same shape, turned.

Both drawings are the same square. Counting corners on each gives four and four:

$$\boxed{4 = 4}$$

The two numbers are equal, and that is the whole lesson on this sheet: turning a shape changes nothing you can count. Same number of sides, same number of corners, same name — **square**.
The sentence to draw out of the child: something like "it's still a square because it still has four sides and four corners." If they say "because it's the same one just turned," that is also correct and arguably better — ask them how they would prove it to someone who did not watch you turn it, and steer them to counting.

8. Your turn to draw. (Open drawing — check it with the child.)

There is no single right answer. Anything with three straight sides and three corners counts, as long as it is not resting flat on a horizontal bottom edge: a triangle standing on its point, one tipped over onto a side, one pointing sideways.
Check two things:

- **The drawing.** Three straight sides that meet, and a closed shape. Wobbly lines are fine at this age; a gap where two sides should meet is worth pointing out gently, because a shape has to be closed.
- **The reason.** Full credit is a reason that names something countable — "it has three sides" or "it has three corners". A reason that names the turn ("because I drew it like a triangle") is the one to push on: ask "how would you show me?" and count together.

If the child draws a triangle sitting flat after all, do not mark it wrong — ask them to turn the paper and look at what they have. It is the same triangle, which is the answer to the question they were actually asked.
Open drawing and explanation — $\boxed{\text{open response}}$.